

Dilawar Singh

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Rust/C++ systems engineer with depth across kernel development (eBPF, Windows WDK) and high-performance async APIs. Built and shipped Shepherd — a cross-platform endpoint security product — from sensors to cloud, leading a team of 15. PhD in computational neuroscience (NCBS); FOSS contributor with work published in eLife, Bioinformatics, and on crates.io.

Skills

- **Systems & Low-Level:** Linux kernel modules, eBPF, Windows Kernel Minifilter (WDK), endpoint security, real-time systems, Win32/POSIX APIs, IPC, SDK development (Rust, C++), GUI (Qt, ImGui), CLI tooling
- **Languages:** Rust, C, C++17/20, Python (expert); Java, TypeScript, JavaScript, Nim, Haskell (intermediate); VHDL, Verilog, SystemC, BlueSpec (familiar)
- **Backend & SaaS:** Microservices and distributed systems; high-performance gRPC/REST APIs (Axum, Actix, FastAPI); PostgreSQL, DuckDB; job queues; event-driven systems; data parsing, validation, and lifecycle management
- **Observability & Infrastructure:** OpenTelemetry, Grafana; Docker, QEMU; CI/CD (GitLab, Jenkins); build systems; packaging (OBS, NSIS); Ansible, Pulumi
- **Other:** Systems architecture, parser development, technical literature review, team leadership and hiring

Work Experience

Independent Consultant — Self-Employed, Bengaluru

Mar 2026 – Present

- Designing and developing scalable microservices supporting a virtual assistant platform for an automotive control system (Rust, C, Qualcomm SoC)
- Designing and overseeing development of high-performance APIs (Rust, AMQP) for a cyber lab running near-real-time experiments on highly correlated, diverse datasets (audio, video, sensor events)

Senior Engineer — Dagnosis, Bengaluru

Jan 2025 – Feb 2026

- Co-developed the sensor data ingestion layer in Rust (Poem, Tokio), adopting Protocol Buffers as the wire format for sensor event streams and aggregating ~30 GB/session and ~10,000 events/session into a single operational control plane
- Designed a unified session model and sensor registry to normalize data across heterogeneous sensor types, enabling consistent querying and incident response
- Instrumented the platform with OpenTelemetry and deployed SignOz for distributed tracing; set up and operated self-hosted infrastructure (GitLab, PostgreSQL, CI/CD runners), including access control, backups, and deployment policies
- Established risk-focused planning and sprint rituals that reduced invisible work, clarified ownership, and gave leadership visibility into team throughput
- Built hiring pipelines and hired 5 engineers, scaling the team from 6 to 11 in 3 months

Cofounder & CTO — Subconscious Compute, Bengaluru

Dec 2019 – Dec 2024

- Wrote Windows kernel minifilter (WDK, C++/Rust) and Linux eBPF sensors generating live telemetry for the cross-platform threat detection and compliance engine, deployed across customer endpoints
- Authored the Windows port in Rust, including unsafe Win32 API bindings and FFI interop with existing C++ components
- Hired and led a team of 15 engineers to build Shepherd, an automated endpoint compliance and security platform, and authored pitch decks that secured accelerator placements

Research Fellow — NCBS Bangalore & GSoC Mentor

2016 – 2019

- Added high-performance numerical solvers (C++, Boost, GNU Scientific Library) to the [MOOSE simulator](#), including a [standalone LSODA solver](#), enabling it to solve multi-scale neural simulation problems
- Wrote Python bindings (pybind11) and created packages for [MOOSE](#) and [Smoldyn](#), significantly improving usability and adoption by the scientific community
- Mentored GSoC contributors on GPU/CUDA-based acceleration across multiple cycles

Firmware Engineer — Kritikal Solutions, Noida

Jul 2009 – Jun 2010

- Implemented a Kalman-filter-based image stabilization system for professional movie cameras (C++, multi-threaded RTOS)

Projects

- Published Rust crates: [simple_accumulator](#) (online statistics via Welford's algorithm, 40k+ downloads), [query-wmi](#) (unsafe Windows COM/WMI bindings), [job-dispatcher](#) (async job dispatch via Tokio)
- Built [fingerprint-sensor-rust-tenesy3.2](#), a bare-metal Rust driver for a biometric fingerprint sensor on the Teensy 3.2 (ARM Cortex-M4)
- Published Python packages: [PlotDigitizer](#) (164 stars — plot image data extractor via OpenCV), [eye-blink-detector](#) (real-time blink detection), [animal-tracking](#) (object tracking for behavioural experiments), [SerialScope](#) (serial-port oscilloscope)

Education

- **PhD, Computational Neuroscience** — NCBS Bangalore (TIFR Mumbai), 2019
- **PhD, Digital Systems** (withdrawn) — IIT Bombay, 2011–2013
- **M.Tech, Electrical Engineering (VLSI)** — IIT Bombay, 2009
- **B.Tech, Instrumentation & Control Engineering** — Dr. MGR ERI Chennai, 2007

Publications

[Full list on Google Scholar](#)

- [Subunit exchange enhances information retention by CaMKII in dendritic spines](#), *eLife*, 2018
- [Python interfaces for the Smoldyn simulator](#), *Bioinformatics*, 2022
- [BioSimulators: a central registry of simulation engines](#), *Nucleic Acids Research (NAR)*, 2022